

IIRSM Course Approval — Full Submission Brief

eLearning Application (No Trainer) | Chainsaw Maintenance & Cross Cutting | Ground Level | Version 1.4

SECTION 1 | Course Identity & Operational Specification

Official Course Title	Chainsaw Maintenance & Cross Cutting (Ground Level Framework)
Course Format	eLearning — Progressive Web Application (PWA) combined with an integrated printed manual; no trainer required
Publisher	Overleaf Publishers Ltd
Course Author / Content Developer	David J Daniel
Copyright	Copyright © 2026 Overleaf Publishers Ltd. All rights reserved. First Edition: July 2026
Platform URL	chainsawcourses.com
Contact Email	info@chainsawcourses.com
Postal Address	69 Newstead Avenue, Orpington, BR6 9RW
NOS Alignment	Independently mapped to UK National Occupational Standards (NOS) for Chainsaw Operations
NPTC Alignment	Assessment criteria independently mapped to City & Guilds NPTC 0039-20 unit parameters for theoretical reference and CPD purposes only
CPD Points Awarded	5 Verifiable CPD Points
Guided Learning Hours (GLH)	4 Hours — PWA online modules, video content & gated knowledge checks
Directed Online Assessment	2 Hours — formative module quizzes + 45-question randomised multiple-choice final examination
Independent Self-Study	4 Hours — reading The Chainsaw Manual & risk assessment exercises
Total Qualification Time (TQT)	10 Hours total
Minimum Pass Threshold	80% or higher on the summative examination — unlimited resit attempts permitted
Target Learner Profile	Commercial chainsaw operators, forestry workers, arborists, estate and grounds teams, and landscape professionals
Course Version	Version 1.4 — submitted for initial IIRSM eLearning Course Approval (1 course)
Approval Category	New to IIRSM Course Approval — eLearning (no trainer)

SECTION 2 | Developer & Author Credentials

IIRSM requires full details of who developed the course and their level of knowledge and skills to do so. The following provides a complete account of the author's qualifications, experience, and subject-matter expertise relevant to this submission.

Author Name	David J Daniel
Role	Course Author, Content Developer & Platform Publisher
Organisation	Overleaf Publishers Ltd
Chainsaw Qualification	City & Guilds NPTC 0039-20 — Chainsaw Maintenance and Cross Cutting (certificated)
First Aid Certification	Current First Aid at Work certificate held
Industry Experience	Over 27 years in the arboricultural and forestry industry. Worked as an arboricultural contract climber for numerous large and small companies throughout this period, gaining extensive hands-on operational experience across a wide range of site conditions and chainsaw applications.
Assessor & Instructor Qualifications	City & Guilds NPTC Assessor and LANTRA Instructor and Assessor for 8 years — covering practical chainsaw assessment and vocational training delivery to industry standard.
Published Author	Author of 'The Chainsaw Manual' (Version 1.1, July 2026) — currently sold as a standalone physical learning aid to various colleges and training providers across the UK. The manual underpins the theoretical content of this eLearning course.
Subject Research	Course content developed with reference to current UK HSE guidance, NPTC 0039-20 unit standards, and the Overleaf Chainsaw Manual (published reference text)
Platform Development	Full-stack eLearning platform designed, developed and operated by author — Progressive Web Application with device-locked access, video streaming, and automated assessment
CPD Status	Author engages in ongoing professional development in chainsaw safety and vocational eLearning design

SECTION 3 | Learning Outcomes & Assessment Criteria

The course is structured across video modules, each mapped to a discrete learning outcome (approximately 90 minutes of video content in total). Learners must complete each module video in full and achieve 80% or higher on the associated module quiz before the next module unlocks. A final summative examination of 45 randomised questions is required for certification.

Learning Outcome	Description & Assessment Criteria Coverage
LO1 Legal Framework & Personal Safety (Unit 1)	Describe the statutory legal framework and personal safety requirements governing chainsaw operations: HSWA employer/employee obligations; PUWER equipment parameters; COSHH control tracking for fuels and lubricants; CE/UKCA class markings for helmets, hearing protection, gloves, and Type A/C protective trousers (AC 1.1–1.4).
LO2 Hazard Evaluation & Emergency Protocols (Unit 2)	Evaluate environmental hazards, risk metrics, and implement emergency protocols: 5-step site-specific risk assessment documenting ground hazards, pedestrian proximity, and structural vulnerabilities; emergency communication and extraction mapping with grid references and trauma kit deployment; bio-security cleaning controls for invasive arboreal pathogens (AC 2.1–2.3).
LO3 Chainsaw Architecture & Safety Features (Unit 2)	Analyse the mechanical differences, design attributes, and safety features of internal combustion and battery-powered chainsaws: 2-stroke combustion cycle and 50:1 fuel-to-oil ratio; battery thermal runaway risks; identification and function of the 10 core safety features across front, centre, and rear chainsaw architecture (AC 3.1–3.3).
LO4 Diagnostic, Servicing & Maintenance (Unit 2)	Describe the diagnostic, servicing, and maintenance procedures required to sustain cutting-assembly integrity: air filter cleaning and spark plug electrode colour interpretation; Rim vs. Spur sprocket differences; guidebar wear conditions (burring, rail splaying, thermal bluing); chain pitch, gauge, tooth shapes (Full-Chisel vs. Semi-Chisel), and filing profiles (AC 4.1–4.4).
LO5 Pre-Use Verification & Startup (Unit 3)	Implement safe pre-use verification protocols and startup methodologies: cold start floor-anchor vs. warm start knee-clamp procedures for Husqvarna and Stihl platforms; 4-point dynamic check assessing chain brake engagement, oil dispersion flow, chain creep at tick-over, and off-switch motor cut (AC 5.1–5.2).

LO6 | Cutting Mechanics: Tension & Compression (Unit 3)

Apply mechanical principles to resolve tension and compression forces during timber cutting: log setup analysis; pulling vs. pushing chain physics; kickback zone definition and safe plunge bore entry; cut sequences for standard logs, oversized timber, and extreme-tension timber using Toast Rack and reduction sink techniques (AC 6.1–6.4).

SECTION 4 | Course Delivery & Technical Specification

Delivery Platform	Progressive Web Application (PWA) — accessible via modern web browsers on desktop, tablet, and mobile
Access Control	Single-use device-locked activation code; code bonds to first device on activation — prevents sharing
Video Hosting	Vimeo Pro — encrypted streaming; watermark overlays display learner name and email dynamically
Assessment Engine	Server-side randomised question bank; results stored in PostgreSQL database with full audit trail
Digital Waiver	Touch/mouse signature capture on first login — legally binding acknowledgement of course terms
Certificate Issuance	Automated PDF certificate issued on passing the final examination — emailed to learner
Data Security	All data encrypted in transit (TLS 1.3); hosted on Replit deployment infrastructure (EU region)
Accessibility	WCAG 2.1 AA compliant colour contrast; keyboard navigable; screen-reader compatible markup
Browser Support	Chrome 120+, Safari 16+, Firefox 120+, Edge 120+
Companion Manual	The Chainsaw Manual by Overleaf Publishers Ltd — Version 1.1 (July 2026); printed; supplied separately; referenced and QR-code linked throughout the digital course

SECTION 5 | Quality Assurance & Review Process

Content accuracy is assured through the following quality control measures:

- All module content cross-referenced against current HSE Chainsaw guidance (HSE AFAG series) and City & Guilds NPTC 0039-20 unit specifications.
- Quiz questions reviewed by the author against published assessment criteria and common examiner focus areas.
- Platform functionality tested across all supported browsers and device types prior to launch.
- Annual content review scheduled — first review due July 2027.
- Learner feedback captured post-examination and monitored for accuracy and relevance signals.
- Any changes to UK legislation or HSE guidance will trigger an unscheduled content review within 30 days of publication.